CHORDIQ + DOMUS

Faster engineering simulation, without losing the physics.

We turn industrial simulations and processes into fast, physics-based, verifiable AI models, checked against the reference simulation or experiment. ChordIQ builds the Physics AI technology; Domus takes it into industrial engineering operations.

THE PROBLEM

Simulation (CFD, finite elements) is accurate, but can be too slow and expensive to run for every design sweep. Generic AI is fast, but can ignore the physics of the process and return a plausible, wrong answer. We combine both: **physics constrains, data calibrates, validation decides.**

WHAT WE DO

Simulation acceleration — focus

CFD with OpenFOAM and surrogate models that answer in seconds what used to take hours, checked against the real simulation.

Forecasting and monitoring

Thermal models, soft sensors and digital twins: they estimate what the plant does not measure directly.

Engineering optimization

Compare geometries and operating conditions before deciding, with every result re-checked against the physical model.

MEASURED VALIDATIONS

1.05–1.49× simulated vs. measured erosion, 90° bend

OpenFOAM CFD against a published experiment (Vieira et al., 2016), 3 tests, within the factor of 2 accepted by industry. AI model on top with 1.0% pressure-drop error on unseen cases. 32-simulation campaign in 23 min.

227,059× faster than the 3D simulation, on a heat sink

Against 3D FEA (FEniCS) on 7 geometries; mean temperature error 4.46% (worst case 7.35%). The speed-up factor varies with machine load during the test; the error is deterministic.

–18.2% drag on an airfoil, target lift kept (0.04%)

104 optimizer iterations, each point recomputed against the physical model and not only against the network, within manufacturing limits.

Limits: these figures hold for the problems and test conditions described. We do not promise the same speed-up or accuracy on your case without first assessing your problem, geometry and data.

HOW WE WORK

1. **Diagnosis under NDA, free of charge:** we understand the problem and tell you whether there is a clear gain.
2. **Proof of feasibility (fixed price):** a quick answer to one technical question.
3. **Engineering pilot:** scope and success metric agreed in writing before the first line of code.
4. **Industrial deployment:** tool in operation, with follow-up.

IN THE PILOT

- **You bring:** the problem and metric on one page; a sample of data or geometry (can be anonymized); a technical person to validate.
- **We deliver:** a working model on your case, compared with your current method, with gain, limits and uncertainty.
- **Terms:** NDA before any data.

Bring us your hardest simulation.
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